Bryce Campbell

CS 499 Milestone Two Narrative

The artifact I selected for enhancement is an Android-based weight-tracking application originally developed as the final project for CS-360: Mobile Architecture & Programming. The app allows users to create an account, log their weight, set a goal weight, and track progress over time. I chose this artifact because it represents a foundational mobile application that demonstrates my early understanding of Android development, UI design, and database integration. It also offered substantial opportunities for improvement, especially through the addition of new features such as a calorie goal calculator and refinements to the underlying code structure.

I selected this artifact for my ePortfolio because it showcases my ability to work with Android UI components, SQLite databases, and multi-activity navigation, all essential skills for mobile software development. The enhancements I implemented required me to diagnose and resolve real-world issues such as null pointer exceptions, incorrect database schema definitions, mismatched widget IDs, and missing intent extras. These challenges highlight my ability to troubleshoot, iterate, and improve an existing codebase, which is a critical competency in professional software engineering.

Although I planned in Module One to meet several course outcomes through multiple enhancements, I ultimately focused on implementing the calorie goal calculator due to the complexity of the issues encountered. The debugging process was far more extensive than anticipated. For example, the calorie calculator initially crashed due to a null spinner reference, caused by variable shadowing in the Java code. Later, the dashboard failed to display the saved calorie goal because the database method responsible for saving the goal omitted the username field, resulting in queries returning no matching rows. These issues required careful analysis of stack traces, method signatures, and database interactions. Because of the time spent resolving these problems, I was unable to complete all originally planned enhancements, and I have updated my outcome-coverage plan accordingly to focus on depth rather than breadth.

Enhancing this artifact taught me several important lessons about software development. First, I learned the importance of clear code organization and documentation. The original project lacked comments and consistent structure, which made it difficult to locate the correct methods and UI elements when integrating the new feature. This slowed down the enhancement process and increased the likelihood of introducing new bugs. Second, I gained a deeper appreciation for understanding Android widgets such as RadioGroups and Spinners. Misunderstanding how these components handle events and data led to several crashes that required revisiting Android documentation and adjusting my implementation approach.

Finally, the debugging process itself was a significant learning experience. Many of the crashes stemmed from small but impactful issues, such as mismatched IDs between XML and Java, missing intent extras, or incorrect SQL column names. Tracking these down required patience, careful reading of Logcat output, and a systematic approach to testing each fix. This experience strengthened my troubleshooting skills and reinforced the importance of incremental testing and validation when enhancing an existing application.

Overall, improving this artifact allowed me to demonstrate growth in mobile development, problem-solving, and software engineering practices. The enhanced version of the app is more functional, more robust, and better aligned with real-world expectations for maintainable code. It now serves as a stronger representation of my abilities within my ePortfolio.

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CS 499: Milestone 3 Narrative

The artifact I selected for enhancement is the workout-planning feature of my Android-based fitness application, originally developed as part of my CS360 Mobile Architecture & Programming coursework. This component of the application allows users to create custom workouts, assign them to specific days of the week, and visually track their weekly exercise plan. The enhanced version of the artifact was created during the current term as part of my CS499 capstone work. I chose this artifact because it represents a natural extension of the weight-tracking system I improved earlier in the program and demonstrates my growth in mobile development, algorithms, and data-structure design. Much like the earlier milestone, this enhancement offered substantial opportunities to refine both the user experience and the underlying logic of the application. The app allows users to create an account, log their weight, set a goal weight, and track progress over time, and the workout planner builds on that foundation by expanding the app into a more comprehensive fitness-tracking tool.

I selected this artifact for inclusion in my ePortfolio because it showcases several advanced skills related to algorithms and data structures. First, the enhancement required designing and implementing a new relational database structure, including a dedicated workouts table and a workout-plan table that maps specific workouts to specific days. This structure demonstrates my ability to model real-world relationships using normalized data and efficient lookup algorithms. Second, the artifact incorporates dynamic UI generation, where exercise options, category indicators, and planner rows are created programmatically based on database content. This required careful algorithmic reasoning to ensure that UI elements were generated consistently and updated correctly when the underlying data changed. Third, I implemented a constraint-enforcement algorithm that ensures users can only create one workout per category. This required detecting existing entries, deleting outdated ones, and inserting updated records, an approach that demonstrates my ability to maintain data integrity through controlled overwrite logic. Finally, the addition of color-coded category indicators and nameplate-style UI elements reflects my ability to integrate algorithmic mapping between categories and visual attributes, improving both usability and clarity.

The artifact was significantly improved during the enhancement process. Originally, the workout planner lacked the ability to assign workouts to specific days, did not enforce uniqueness of workouts by category, and did not visually display assigned workouts. Through the enhancement, I added a new database table for day assignments, implemented lookup algorithms to retrieve assigned workouts, and created a dynamic, color-coded indicator system that displays each day's workout category in a visually appealing nameplate format. These improvements not only expanded the functionality of the application but also strengthened its architectural consistency and user experience. Compared to the earlier milestone, where I wrote that the debugging process was far more extensive than anticipated, this enhancement required even deeper debugging and architectural refinement, particularly in aligning database schema changes with dynamic UI updates.

I met the course outcomes I planned to address in Module One, particularly those related to algorithms, data structures, and software design. The enhancements required designing new relational structures, implementing multi-step retrieval algorithms, and integrating dynamic UI logic that responds to database state. These tasks align directly with the outcomes related to developing efficient, maintainable, and scalable software solutions. I do not need to update my outcome-coverage plan, as the enhancements fully address the intended learning goals and demonstrate mastery of the targeted competencies.

Reflecting on the enhancement process, I learned several important lessons about software engineering. First, I gained a deeper appreciation for the importance of database schema planning. Adding new tables and enforcing category uniqueness required careful coordination between SQL definitions, Java methods, and UI logic. Second, I learned how critical it is to maintain synchronization between database queries and object models. At one point, a mismatch between cursor column indices and the Workout model caused a crash similar to issues I described in my earlier narrative, such as incorrect SQL column names and missing intent extras. Debugging these issues reinforced the importance of incremental testing and careful validation. Third, I learned how dynamic UI generation can introduce subtle layout challenges, such as column alignment and weight distribution. Solving these required both algorithmic reasoning and an understanding of Android's layout system. Finally, the process of adding color-coded nameplates taught me how visual design and algorithmic logic intersect, requiring thoughtful mapping between categories, colors, and drawable resources.

Overall, enhancing this artifact allowed me to demonstrate growth in mobile development, algorithmic thinking, and data-structure design. The improved workout planner is more functional, visually polished, and architecturally sound. It now serves as a strong representation of my abilities in mobile software engineering and is well-suited for inclusion in my ePortfolio.

Bryce Campbell

CS 499: Milestone 4 Narrative

The artifact I selected for enhancement in Milestone Three is the workout-planning and workout-creation system within my Android-based fitness application. This component was originally developed as part of my CS 360: Mobile Architecture and Programming final project, which centered on a mobile app that allowed users to log their weight and track progress toward a goal. In my Module One enhancement plan, I identified this project as the foundation for all three categories of improvement—software design, algorithms and data structures, and databases—because it offered substantial opportunities for expansion into a more holistic fitness-tracking tool. The database enhancement completed in this milestone focused specifically on improving the underlying data structures that support the workout planner, including the creation of new relational tables, enforcement of data-integrity constraints, and refinement of SQL queries and schema definitions. These enhancements were implemented during the current term as part of my CS 499 capstone work.

I selected this artifact for inclusion in my ePortfolio because it demonstrates my ability to design and implement database solutions that support complex application features. In my Module One plan, I proposed expanding the existing SQLite schema by adding multiple relational tables for exercises, workout logs, and BMR-related profile data, as well as integrating advanced SQL concepts such as joins, aggregate functions, and multitable queries. The enhancements completed in this milestone align directly with those goals. Specifically, I added a dedicated workouts table to store user-created workouts and a workout_plan table to map those workouts to specific days of the week. This required designing normalized relational structures and implementing foreign-key-like relationships through consistent identifiers. Additionally, I implemented a constraint-enforcement algorithm that ensures users can only create one workout per category. This overwrite logic required detecting existing entries, deleting outdated ones, and inserting updated records, demonstrating my ability to manipulate stored data through controlled and efficient SQL operations. These improvements showcase my skills in database design, data-structure planning, and algorithmic reasoning, fulfilling the enhancement category for databases.

The artifact was significantly improved through the enhancement process. Before these changes, the workout planner lacked the ability to store workouts in a structured relational format, did not enforce uniqueness of workouts by category, and had no mechanism for assigning workouts to specific days. The enhancements introduced new SQL tables, improved schema definitions, and added multi-step retrieval algorithms that allow the application to fetch assigned workouts and display them dynamically. I also addressed several structural issues that emerged during development, such as missing tables, mismatched cursor indices, and schema-version conflicts. These issues were similar to the challenges I described in my Milestone Two narrative, where I noted that the original project required resolving incorrect database schema definitions and incorrect SQL column names, and the debugging process in this milestone continued to reinforce the importance of careful schema planning and validation. The final result is a more robust, secure, and maintainable database foundation that supports the expanded functionality of the workout planner.

I met the course outcomes I planned to address in Module One, particularly those related to database design, data manipulation, and secure data handling. The enhancements required implementing innovative database techniques, such as enforcing category-level uniqueness, designing relational structures for day assignments, and integrating dynamic SQL queries that respond to user actions. These tasks align directly with the outcomes related to developing efficient and secure database solutions. I do not need to update my outcome-coverage plan, as the enhancements fully address the intended learning goals and demonstrate mastery of the targeted competencies.

Reflecting on the enhancement process, I learned several important lessons about database engineering. First, I gained a deeper appreciation for the importance of schema evolution and version control. Adding new tables required updating the database version and ensuring that the onUpgrade() method properly recreated all necessary structures. Second, I learned how critical it is to maintain synchronization between SQL queries and the object models used in the application. At one point, a mismatch between cursor column indices and the Workout model caused a crash similar to issues I described in my earlier milestone, such as incorrect SQL column names and missing intent. Debugging these issues reinforced the importance of incremental testing and careful validation. Third, I learned how enforcing data-integrity constraints, such as limiting users to one workout per category, requires thoughtful algorithmic design and a clear understanding of how relational data should behave. Finally, the process of integrating database logic with dynamic UI elements taught me how closely database design and user experience are connected, especially when the UI must reflect real-time changes in stored data.

Overall, enhancing this artifact allowed me to demonstrate growth in mobile development, algorithmic thinking, and data-structure design. The improved workout planner is more functional, visually polished, and architecturally sound. It now serves as a strong representation of my abilities in mobile software engineering and is well-suited for inclusion in my ePortfolio.